

DL for ENG

LECTURE 8.1

Stationary Heat

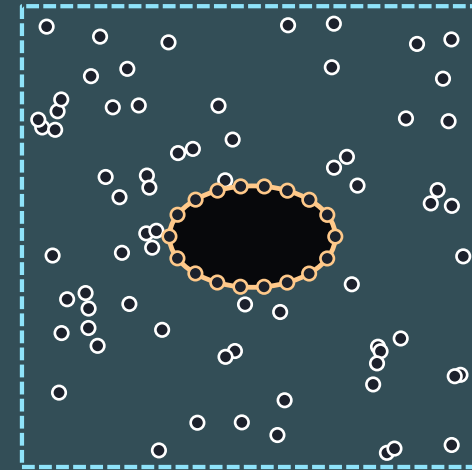
Remus Teodorescu

*With support from Research Assistant Noman Khan -
nomank@energy.aau.dk*

Aalborg University · ret@et.aau.dk

Deep Learning for Engineering · 2026

CONDUCTION · GEOMETRY · THE PLATE



Steady conduction is Poisson's equation with units: Fourier's law, a conductivity, and walls held at a temperature, a flux or a convection. On a plate with a cooling hole the hole is built into the network by its level set, and the answer is checked by a flux balance.

Learning Objectives

SECTIONS, IN ORDER

- 1 **Conduction** — Fourier's law, variable k , the thermal walls, scaling
- 2 **Geometry** — sampling a domain with a hole, its multiplier, its normal
- 3 **Plate with a hole** — verify, solve, weight, and balance the heat

READ, AND WHAT EACH ANSWERS

- | | |
|--------------------|--|
| Liu ch. 2.4 | what does a real thermal problem need? |
| Liu ch. 3 | how is a domain with a hole handled? |
| Liu ch. 4.6 | how is the answer checked? |

Recap

1 – 5 · STEADY

- 1 What decides the type of a second-order equation, and which conditions does each type need? *Liu ch. 2; Ex_07.2 notebook 04*
- 2 What does Poisson's equation describe, why does it need a condition on every wall, and what goes wrong with a flux on all of them? *Liu ch. 2.4; Ex_07.2 notebook 04*
- 3 How is a Helmholtz network trained: why does it find $u \equiv 0$, and which two safeguards exclude it? *Liu ch. 6.2, 6.4.3; Ex_07.2 notebook 04*
- 4 How does a steady problem differ from one in time: which input and which conditions does it do without? *Liu ch. 2, 4.1*
- 5 What are Poisson and Helmholtz used for, and how does orthogonality find the modes after the first? *Liu ch. 6.4 – 6.5*

6 – 10 · IN TIME

- 6 What does the heat equation add to Poisson, and what does its type do to an error made early? *Liu ch. 4.1; Ex_07.2 notebook 03*
- 7 How is the heat equation's initial field imposed, soft or built in, and why is it the term a PINN most often loses? *Liu ch. 4.1 – 4.2; Ex_07.2 notebooks 01 – 02*
- 8 How is a wave network tuned: what does a sine activation buy, and how must it be compared with tanh? *Liu ch. 5.7.4; Ex_07.2 notebook 03*
- 9 How does the wave equation differ from the heat equation, and why does it need two initial conditions? *Liu ch. 5.1 – 5.2; Ex_07.2 notebook 03*
- 10 What are heat and wave used for, and what did the panel without its velocity condition show about a low loss? *Liu ch. 5.1 – 5.2; Ex_07.2 notebook 04*

Steady Conduction

Steady conduction — heat flows down the temperature gradient, $q = -k\nabla T$ (Fourier's law), and in steady state what leaves a small volume equals what is generated in it, $\nabla \cdot q = Q$. Together they give Poisson's equation for the temperature, with Q/k as its source.

FOURIER'S LAW IN AN ENERGY BALANCE

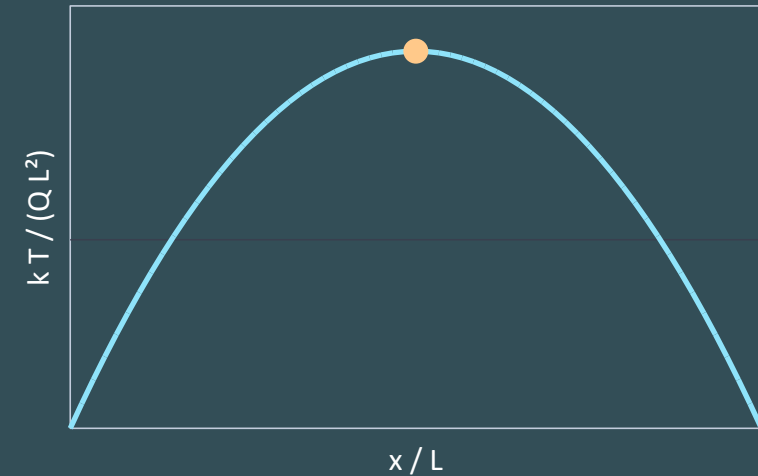
$$q = -k \nabla T, \quad \nabla \cdot q = Q \qquad T_{xx} + T_{yy} + \frac{Q}{k} = 0$$

- **k** — the conductivity, in W/(m K)
- **Q** — the heat generated, in W/m³

CONDUCTION IN PYTORCH

```
T = net(xy)                # the temperature, in kelvin
q = -k * grad(T, xy)        # Fourier's law; grad = autograd
F = div(q, xy) - Q          # the energy balance, zero inside
```

A WALL THAT GENERATES HEAT



Both faces held at T_0 . The profile is a parabola, and at the amber peak the gradient, and so the flux, is zero: heat leaves through both faces.

Variable Conductivity

Variable conductivity — when k changes with position it stays inside the divergence: $\nabla \cdot (k \nabla T) + Q = 0$. Across an interface between two materials the gradient jumps and the normal flux does not, so the divergence must not be expanded; autograd differentiates $k \nabla T$ as written.

K INSIDE THE DIVERGENCE

$$\nabla \cdot (k(\mathbf{x}) \nabla T) + Q(\mathbf{x}) = 0$$

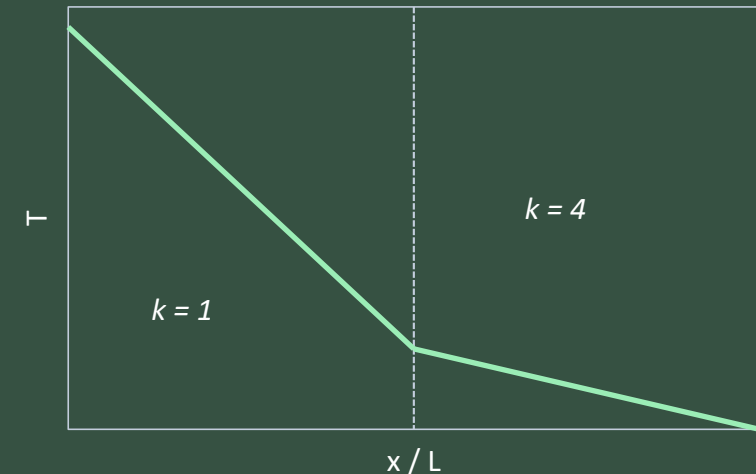
the conservative form: the flux $k \nabla T$ is differentiated as one quantity

- **at an interface** — the slope jumps, the flux does not
- **in code** — differentiate the product, not its parts

VARIABLE k IN PYTORCH

```
k = k_field(xy)           # the conductivity at each point
q = -k * grad(T, xy)       # the flux, one quantity
F = div(q, xy) - Q         # differentiated as written
```

TWO MATERIALS, ONE INTERFACE



Left face at 1, right face at 0, no generation. The slope falls fourfold at the interface; the flux $k dT/dx$ is the same on both sides.

Thermal Boundary Conditions

Thermal boundary conditions — three ways to fix a wall: a temperature (Dirichlet), a heat flux (Neumann), or convection to a fluid at T_∞ with film coefficient h (Robin). A temperature can be built into the network; a flux or convection condition stays a loss term.

A TEMPERATURE, A FLUX, A CONVECTION

Dirichlet $T = \bar{T}$ on $\partial\Omega_D$

Neumann $-k \frac{\partial T}{\partial n} = \bar{q}$ on $\partial\Omega_N$

Robin $-k \frac{\partial T}{\partial n} = h(T - T_\infty)$

THE WALLS IN PYTORCH

```
B_D = T(x_D) - T_bar           # Dirichlet: the temperature
B_N = -k * dTdn(x_N) - q_bar     # Neumann: the heat flux
B_R = -k * dTdn(x_R) - h * (T(x_R) - T_inf) # Robin: flux tied to T
loss = L_pde + w_N * B_N.pow(2).mean() + w_R * B_R.pow(2).mean() # weighted
```

WHAT EACH FIXES, AND HOW

	fixes	enforced
Dirichlet	the temperature	built in
Neumann	the heat flux	a loss term
Robin	flux and value	a loss term

The Biot number $Bi = hL/k$ compares the film with the conduction. For a 1 mm silicon device in still air, $h = 25 \text{ W/(m}^2 \text{ K)}$ and $k = 150 \text{ W/(m K)}$, Bi is 1.7×10^{-4} : the convection wall is nearly insulated.

Well-Posedness

Compatibility condition — a problem with a flux condition on every wall has a solution only if the heat out through the walls equals the heat generated, and then only up to a constant: any shift of T fits the same equation and walls. A PINN picks one without warning.

HEAT OUT EQUALS HEAT GENERATED

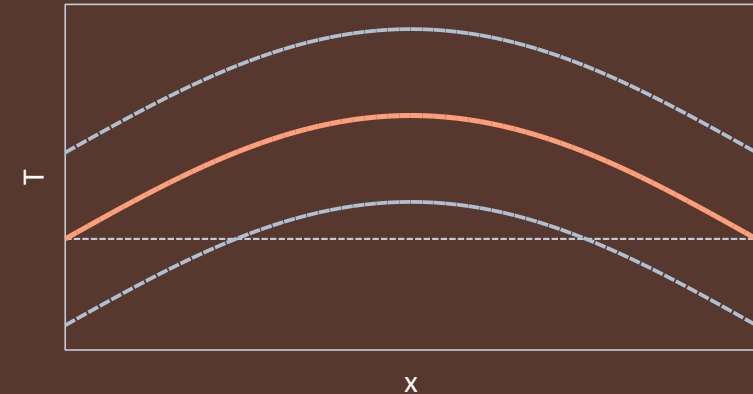
$$\oint_{\partial\Omega} \bar{q} ds = \int_{\Omega} Q d\Omega$$

the flux through the whole boundary against the heat generated inside

- **one value pinned** — fixes the constant
- **one wall at a temperature** — fixes it too

TALK · 2 MIN How would you explain why a problem with only flux conditions has no single solution, in your own words? Then, in the plot on the right, which of the three curves is right, and what would you add to the loss to choose it?

THE SAME FIELD, SHIFTED



FIXING THE LEVEL

```
L_pin = (T(x_ref) - T_ref).pow(2)    # pin one value
L_mean = T(xy).mean().pow(2)         # or fix the mean
```

Non-Dimensional Form

Non-dimensional form — the problem rewritten in scaled variables: lengths divided by the device size L , temperature by a rise $\Delta T = QL^2/k$. Every term is then of order one, which is what the initialisation, the activations and the loss weights assume. Train scaled, and convert back at the end.

SCALE, TRAIN, CONVERT BACK

$$\theta = \frac{T - T_0}{\Delta T}, \quad \xi = \frac{x}{L}, \quad \Delta T = \frac{Q L^2}{k}$$

the scaled temperature θ , the scaled coordinates, the temperature scale

$$\theta_{\xi\xi} + \theta_{\eta\eta} + 1 = 0$$

the same equation with every coefficient equal to one

SCALING IN PYTORCH

```
dT = Q * L**2 / k           # the temperature scale, kelvin
theta = net(xy_m / L)       # train on scaled coordinates
T = T0 + dT * theta         # convert back, in kelvin
```

A SILICON DEVICE, BEFORE AND AFTER

	physical	scaled
length L	1 mm	1
conductivity k	150 W/(m K)	1
source Q	10^8 W/m ³	1
rise ΔT	$Q L^2/k = 0.67$ K	1
film h	25 W/(m ² K)	$Bi = 1.7 \times 10^{-4}$

In metres and kelvin the source term is ten to the eighth and the length ten to the minus third. Scaled, every term is of order one.

Sampling a Real Domain

Rejection sampling — points drawn uniformly over a box around the domain, keeping those the level set places inside, $\phi > 0$. On a curved wall the points are spaced by arc length rather than by angle, so the ends of the hole, where it bends most, get their share.

ONE FORMULA: INSIDE, OUTSIDE, ON THE WALL

$$\varphi(x, y) = \left(\frac{x - x_c}{a}\right)^2 + \left(\frac{y - y_c}{b}\right)^2 - 1$$

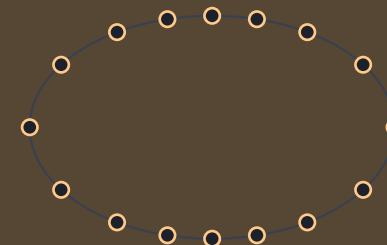
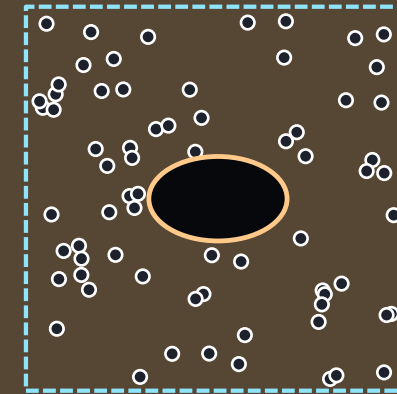
negative in the hole, zero on its wall, positive in the plate

- **inside** — keep a point where $\phi > 0$
- **on the wall** — equal steps of arc length

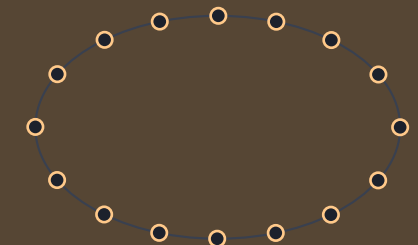
SAMPLING IN PYTORCH

```
xy = torch.rand(4000, 2)           # uniform over the square
xy = xy[phi(xy) > 0][:2000]        # keep the points in the plate
xy_h = ellipse_by_arc_length(160)  # the hole wall, evenly spaced
```

THE PLATE, AND ITS WALL POINTS



equal angle: ends bare



equal arc length

Boundary Multiplier

Boundary multiplier — a function that is zero on a boundary and positive in the domain. Multiplied into the network's output, it makes a value hold there exactly. For an ellipse the level set ϕ is one; for a shape with no formula, a small second network is trained to be one, then frozen.

THE HOLE WALL, BUILT IN

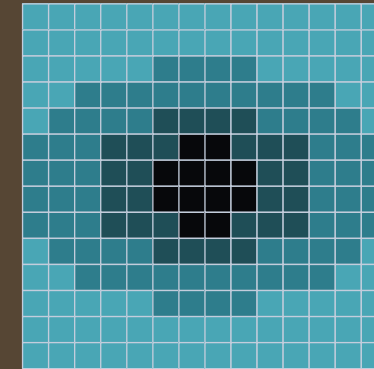
$$T(\mathbf{x}) = \varphi(\mathbf{x}) \mathcal{N}_{\hat{w}}(\mathbf{x})$$

$T = 0$ on the hole for any weights: φ vanishes there

- **a value** — held exactly by a factor
- **a derivative** — cannot be, stays a loss term

TALK · 2 MIN How would you explain what a boundary multiplier does, in your own words? Then, in the figure on the right, where is the multiplier small, and what does that do to the network's gradient near the hole?

ϕ OVER THE PLATE



dark: the hole

THE HOLE IN PYTORCH

```
T = hole_multiplier(xy) * net(xy)  # zero on the hole
loss = F.pow(2).mean() + w_N * L_N  # edges still a term
```

Flux on a Curved Boundary

Normal derivative — the rate of change of T across a wall, $\partial T / \partial n = \nabla T \cdot \mathbf{n}$, where $\mathbf{n}$ is the unit normal pointing out of the material. On a curved wall $\mathbf{n}$ changes with position: it comes from the gradient of the level set, and its sign decides whether heat leaves or enters.

THE NORMAL FROM THE LEVEL SET

$$\frac{\partial T}{\partial n} = \nabla T \cdot \mathbf{n}, \quad \mathbf{n} = -\frac{\nabla \phi}{|\nabla \phi|}$$

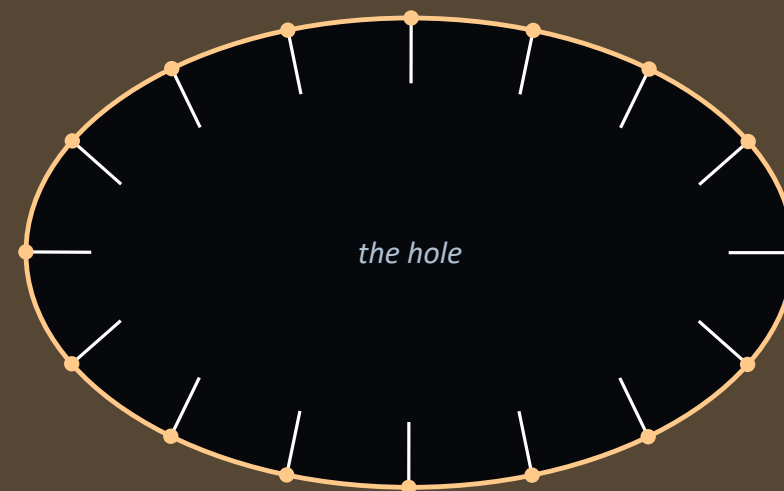
minus: out of the plate is into the hole

- **on an edge** — the normal is a constant
- **on the hole** — it turns with the wall

THE NORMAL IN PYTORCH

```
g = grad(phi(xy_h), xy_h)           # the level set's gradient
n = -g / g.norm(dim=1, keepdim=True) # unit normal, out of the plate
q_n = -k * (grad(T, xy_h) * n).sum(1) # the heat flux through the wall
```

NORMALS ON THE HOLE WALL



Each white stick is $\mathbf{n}$ at an amber point: it turns with the wall and always points out of the plate. A sign error inverts the flux, not the magnitude.

Manufactured Solution

Manufactured solution — a solution chosen in advance, with the source derived from it by differentiation, so the answer is known exactly. Solving that problem checks the residual, the trial solution and the training loop before they meet a problem with no answer to compare against.

CHOOSE THE ANSWER, DERIVE THE SOURCE

$$T = \sin(\pi x)\sin(\pi y) \Rightarrow \frac{Q}{k} = 2\pi^2 \sin(\pi x)\sin(\pi y)$$

zero on all four edges of the unit square, so $x(1-x)y(1-y)$ holds the walls

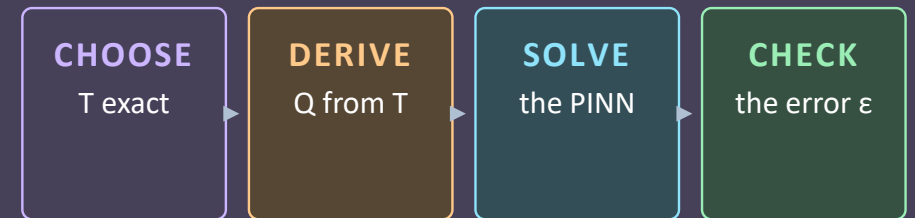
$$\varepsilon = \frac{\|T_{\hat{w}} - T_{\text{exact}}\|_2}{\|T_{\text{exact}}\|_2}$$

the relative error against the answer you chose

THE CHECK IN PYTORCH

```
Q_k = 2 * pi**2 * sin(pi*x) * sin(pi*y)    # the source, from the answer
T = x*(1-x) * y*(1-y) * net(xy)           # the walls held exactly
err = (T - T_exact).norm() / T_exact.norm() # compare with the answer
```

FOUR STEPS BEFORE THE PLATE



what it checks

the residual, the trial, the loop

what it cannot

the geometry of the plate

An error found here costs minutes; the same error found on the plate costs an afternoon.

Heated Plate with a Hole

Heated plate with a hole — the case study of this block: a square plate with uniform heat generation, cooled through an elliptical hole held at $T = 0$, with insulated outer edges, $\partial T / \partial n = 0$. The hole wall is built into the network by its level set; the edges stay a loss term.

ONE BUILT-IN WALL, ONE LOSS TERM

$$\mathcal{L} = \mathcal{L}_{PDE} + w_N \mathcal{L}_{NBC}$$

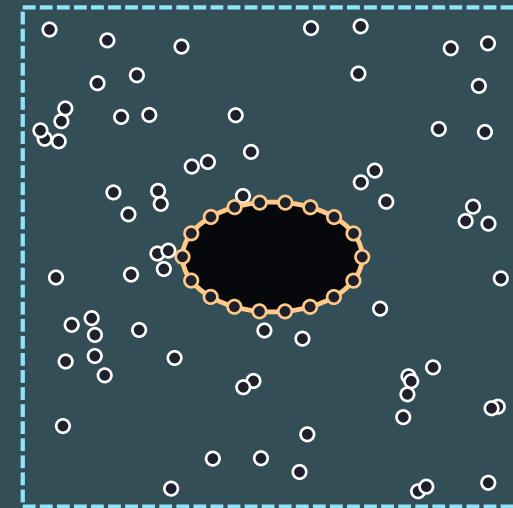
the residual inside, and the insulated edges with their weight

- **the hole** — $T = 0$, by the multiplier
- **the edges** — no flux, a weighted term

THE PLATE IN PYTORCH

```
T = hole_multiplier(xy) * net(xy)          # the hole held at zero
L_N = dTdn(T, xy_edges).pow(2).mean()      # the edges: no flux
loss = (lap(T) + Q_k).pow(2).mean() + w_N * L_N # residual and edges
```

THE PLATE OF EX_08.1



amber wall

$T = 0$, built in

dashed edges

$\partial T / \partial n = 0$, a loss term

Flux Weight

Flux weight — the weight w_N on the insulated-edge term. Too small, and the network ignores the edges and heat leaks through them; too large, and the residual inside pays for a perfect wall. It is chosen by a sweep that watches both terms, not the total loss.

ONE RUN PER WEIGHT, BOTH TERMS WATCHED

$$\mathcal{L} = \mathcal{L}_{PDE} + w_N \mathcal{L}_{NBC}$$

- **the total loss** — hides which term is paying
- **the choice** — a trade, argued from both

THE SWEEP IN PYTORCH

```
for w_N in [0.1, 1, 10, 100]:  
    net = train(w_N)           # one run per weight  
    log(w_N, L_pde(net), L_N(net)) # the same seed, the same points  
                                # both terms, not their sum
```

WHAT EACH END OF THE SWEEP DOES

w_N small

the edges leak heat

w_N large

the residual inside pays

between

argued from both terms

The weight is a rate of exchange between a flux and a residual, which have different units. Scaling changes the rate before any sweep does.

Flux Balance

Flux balance — in steady state the heat leaving through the hole must equal the heat generated in the plate: the perimeter times the mean normal flux, against Q times the area. It is a check with a physical reason to come out right, and it needs no exact solution.

HEAT OUT AGAINST HEAT GENERATED

$$P_h \overline{q_n} = Q A_\Omega$$

P_h the hole's perimeter, A_Ω the plate's area without the hole

- **an exact solution** — not needed
- **the mean flux** — needs evenly spaced points

THE BALANCE IN PYTORCH

```
q_n = -k * (grad(T, xy_h) * n).sum(1)    # flux into the hole
out = q_n.mean() * perimeter              # heat leaving the plate
gen = Q * (1 - pi * a * b)                # heat generated in the plate
```

NOTEBOOK 03, SCALED UNITS

	value
heat generated	0.93780
heat out	0.92438
out short by	1.4 %

The generated heat is Q times the plate's area, $1 - \pi \cdot 0.18 \cdot 0.11$. The heat out is the network's. Whether 1.4 % is small enough is a decision the method cannot make for you.

Advanced Topics

RELATED TOPICS LEFT OUT, AND WHERE THEY MATTER

topic	what it adds	read	where it makes a difference
energy method	an energy, first derivatives only	Liu ch. 2.8	elasticity, structures
triangulated domain	points and weights from a mesh	Liu ch. 3.2	any drawn geometry
convergence study	error against points and size	Liu ch. 4.9.3	a design sign-off
<u>diffusion surrogate</u>	trained on solved cases, then steered	UDL ch. 14 – 18	many boundary conditions
PINN or FEM	when each is the better tool	Liu ch. 2.10	one-off steady solves

WHY THEY ARE LEFT OUT

The plate of Ex_08.1 needs none of them: its hole has a formula, its answer can be checked by a flux balance, and it is solved once. A finite-element solver would solve it too; the method earns its place on the inverse problems of L8.2.

Exercise 8.1

THE NOTEBOOKS, IN ORDER

- **00** — the plate, its hole, its samples, its normals
- **01** — the method, on a manufactured solution
- **02** — the plate with its hole built in
- **03** — the flux weight, and the flux balance
- **04** — compares them, and the report

THE QUESTION EACH NOTEBOOK ANSWERS

- 01** is the method right?
- 02** how hot, and where?
- 03** does the heat balance?

REPORT QUESTION, TO CONCLUDE

The manufactured solution checked the method and the flux balance checked the plate. On a design with no exact solution, which checks would you run, and what result would make you distrust the network?

Key Takeaways

STEADY CONDUCTION

what it is Fourier's law in an energy balance

function T for a given k, Q and walls

train, tune scale first; flux walls weighted

vs Poisson units, k inside, three walls

used for devices, plates, heat sinks

REAL GEOMETRY

what it is a domain with no polynomial factor

function ϕ : inside, multiplier, normal

train, tune arc length on the wall; w_N

vs rectangle n and the factor vary with x

used for plates with holes, device sections

Next — L8.2: the same plate, in time.

Questions

1 – 5 · STEADY CONDUCTION

- 1 What is steady conduction: how do Fourier's law and an energy balance give Poisson's equation, and where does k go when it varies? *Liu ch. 2.4*
- 2 Which three thermal boundary conditions are there, and which of them can be built into the network? *Liu ch. 2.2; Ex_08.1 notebooks 01 – 02*
- 3 How is a device problem trained: why scale it first, and what breaks if you do not? *Liu ch. 4.4; Ex_08.1 notebooks 01, 03*
- 4 How does a problem with only flux conditions differ: when does it have a solution, and what is left undetermined? *Liu ch. 2.2; Ex_08.1 notebooks 02 – 03*
- 5 What is a manufactured solution used for, and why is it the first run on a problem with no reference? *Liu ch. 2.4; Ex_08.1 notebook 01*

6 – 10 · REAL GEOMETRY

- 6 How are points placed on a plate with a hole, and why by arc length rather than angle on its wall? *Liu ch. 3.1 – 3.2; Ex_08.1 notebooks 00, 02*
- 7 What does a boundary multiplier do, and when is it a formula rather than a trained network? *Liu ch. 3.3; Ex_08.1 notebooks 01 – 02*
- 8 How is the flux weight chosen, and what do the flux term and the residual each prefer? *Liu ch. 2.2.3; Ex_08.1 notebook 03*
- 9 How is the flux through a curved wall computed, and what happens if the normal points the wrong way? *Liu ch. 3; Ex_08.1 notebooks 00, 03*
- 10 What does the flux balance check, and would you sign off a design on a mismatch of one or two per cent? *Liu ch. 4.6; Ex_08.1 notebook 03*

NEXT

L8.2 — Transient Heat

The same plate, in time: an initial field, causality, and a diffusivity recovered from sensors.

BEFORE THE NEXT SESSION

read	Liu ch. 3, 4.6
run	Ex_08.1 notebooks 00 to 03
bring	the flux balance, and your w_N
think about	which of your walls can be built in

The steady plate is solved and checked. Next it heats up.